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SARDAR PATEL UNIVERSITY
B.C.A. Semester – IV (NC) Examination, 2022
US04CBCA03 : OPERATING SYSTEM
Friday, 23rd SEPTEMBER 2022

Time: 12 :30 P.M. To 02 :30 P.M.

Marks: 70

Q-1 Multiple Choice Questions.

[10]

1. A newly created process resides in _____ queue.
A. Job B. Ready C. Running D. Waiting
2. _____ scheduling algorithm gives minimal average waiting time.
A. FCFS B. SJF C. Round-robin D. Priority
3. Which of the following is not an operating system structure?
A. Virtual Memory B. Client-server C. Layered System D. Monolithic Structure
4. _____ scheduling algorithm gives minimum Page Faults.
A. FIFO B. Optimal C. LRU D. LIFO
5. In Paging, Physical memory is divided into fixed-size blocks is called _____.
A. Files B. Pages C. Frames D. Algorithm
6. Virtual Memory is commonly implemented by _____.
A. Fragmentation B. Compaction C. Demand Paging D. Backing store
7. _____ buffer has fixed buffer size.
A. Flexible B. Unbounded C. Shared D. Bounded
8. Multiple process solution is also known as _____.
A. Paging Algorithm B. Bankers Algorithm C. Bakery Algorithm D. Virtual Algorithm
9. _____ command is use to change a permission of a file.
A. change B. crmod C. chmod D. man
10. _____ command is use to merge multiple files.
A. merge B. mr C. cp D. paste

Q-2 Fill in the blank and True - False.

[08]

1. Compiler is an example of System Program components of Operating system. [True / False]
2. PCB Stands for _____.
3. Suit-fit is a Memory allocation algorithm. [True / False]
4. A Lazy Swapper is also known as _____.
5. Printer is an example of _____ resource.
6. EXE is not a valid LINUX file system. [True / False]
7. _____ redirection operator is use for input redirection.
8. yy option of vi editor is use to delete a line. [True / False]

Q-3 Give answers of following questions. (Any Ten)

[20]

1. Define Operating system.
2. What is process? List out all process States.
3. Differentiate between **Hard real time** system and **Soft real time** system
4. Explain Virtual memory in short.
5. Explain External Fragmentation in short.
6. Calculate Page faults using FIFO algorithm for following reference string: (Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1
7. When Race conditions arise?
8. Why is LINUX an open source?
9. Justify "Linux is a Secure Operating System".
10. Explain who command in brief.
11. Explain use of mkdir and rmdir command.
12. Explain mv command with all its attributes.

Q-4 Long Questions. (any four)

[32]

1. Explain process state and PCB in brief.
2. Explain Real Time Operating System with Example.
3. Explain Demand Paging in brief
4. Explain Memory allocation techniques in detail.
5. What do you mean by Deadlock? Explain all necessary conditions for occurrence of deadlock.
6. Explain EXT2 File system in details.
7. Explain if and case statement in LINUX.
8. Explain following commands
A. pwd B. rmdir C. who D. ls
